Contact

sbarreto.cardoso@gmail.com

www.linkedin.com/in/saulobarreto (LinkedIn)

Top Skills

Jupyter

Scientific Computing

Testing

Languages

French (Professional Working)

Inglês (Full Professional)

Português (Native or Bilingual)

Honors-Awards

Brazilian Mathematics Olympiad of Public Schools

Brazilian Mathematics Olympiad of Public Schools

Publications

Training and analysis of hyperparameters in neural networks for computer vision applications: A didactic approach

Physically-constrained block-term tensor decomposition for polarimetric image recovery

Using Synthetic Images for Deep Learning Recognition Process on Automatic License Plate Recognition

Décomposition tensorielle de rang faible pour l'imagerie de Stokes polarimétrique

Saulo Barreto

PhD in High-dimensional Data | AI/ML Researcher

Brazil

Summary

PhD in High-Dimensional Data Analysis and Optimization with 10 years of Python expertise spanning Statistical Learning, Machine Learning, Computer Vision, and Data Science.Built across different domains: developed an open-source Python library for high-dimensional image processing, created QR code recognition software for laparoscopic surgery, applied visual pattern recognition to humanoid robot control, and delivered 3D reconstruction and real-time visualization tools for metaverse applications at Bell Labs.Combines strong research foundations with a proven ability to prototype, ship, and explain across academia, industry, and public institutions.

Experience

Vetto AI

Operations Specialist

June 2026 - Present (5 months)

- Project Lead — Annotation Campaign: Led a 40-annotator campaign (Computer Vision domain), overseeing quality, payments, and stakeholder communication

- Research — LLM Evaluation: Led LLM evaluation research (judge design, benchmarks, rubrics, bias/correctness/consistency analysis) with evidence-traceability frameworks.

- Support Team: Closed tickets, built automations, developed dashboards, and organized workflows to streamline support operations.

Independent Contractor

Quality Assurance Specialist

October 2025 - May 2026 (8 months)

Remote

Drove LLM data quality across multiple AI companies, contributing to training, evaluation, and benchmarking initiatives spanning diverse domains.

At Vetto:

- Evaluated 170+ multi-domain datasets (education, health, political) against project specifications, ensuring data integrity and consistency to support LLM training quality.

At Handshake:

- Reviewed and audited 200+ graduate-level mathematics prompts engineered to stress-test state-of-the-art LLMs, evaluating model responses against prior assessments to ensure correctness and logical consistency. Separately led prompt design and rubric development for a commerce-domain AI evaluation project.

At Mercor:

- Designed domain-specific prompts for structured coding tasks, developing corresponding tests and refactoring code to validate outputs. Contributed to a model benchmarking initiative, evaluating state-of-the-art LLMs across complex tasks using a structured criterion framework to assess and compare performance.

Universite de Lorraine
Data Science Instructor | Applied Statistics & Programming
February 2024 - November 2025 (1 year 10 months)
Nancy, Grand Est, France

- Delivered 84 hours of in-person practical sessions using Python and MATLAB for students at different departments and levels: MSc. in NLP, Cognitive Sciences, Law & Economics, and Engineering
- List of courses: Introduction to Data Science | Machine Learning, Probability, Statistics & Data Analysis | Modulation, Coding & Information Theory | Practical Optimization & Neural Networks

CRAN - Research Centre for Automatic Control
PHD Student
October 2022 - November 2025 (3 years 2 months)
Nancy, Grand Est, France

- Developer of a new open-source Python library for processing high-dimensional imaging data (biomedical imaging, radar, etc) using tensors, focusing on scalable performance

- Designed and implemented robust algorithms for large-scale data analysis, integrating linear algebra, optimization, and statistical modeling.

Baeldung
Author & Technical Writer
August 2021 - May 2025 (3 years 10 months)

- Wrote 25+ tutorials and articles on Math for Data Science, Machine Learning, and Programming, collaborating with five editors, committed to making complex topics accessible

Nokia Bell Labs
Research Trainee
January 2022 - July 2022 (7 months)
Budapest, Hungary

- Created 3D reconstructions of real-world scenarios to explore metaverse concepts and applications, including tools for manipulating and transforming point clouds
- Built a web-based visualization platform for interactive and real-time exploration (C++, Three.js, HTML, Python)

University of Debrecen
Developer
May 2021 - December 2021 (8 months)
Debrecen, Hajdú-Bihar, Hungary

- Developed a system to measure vital signals (ECG, EKG, etc.) of pilots, applying Machine Learning to detect high-risk levels and prevent accidents in the Student Proof of Concept program at the University of Debrecen
- Supported a side-project analyzing pilot data collected during the pandemic.

TransPerfect
Data Annotator
March 2021 - April 2021 (2 months)

- Part of a 100+ member team responsible for labeling a large dataset for an OCR project, following strict guidelines for machine learning model training.

Ministério Público Federal

2 years 1 month

IT and Communication Coordination
January 2017 - January 2019 (2 years 1 month)
Greater Brasilia

- Contributed to the production of technical documents for training federal prosecutors and lawyers, participated in workflow optimization meetings, and assisted in monitoring and resolving HelpDesk calls.

Engineering and Architecture Department
January 2018 - May 2018 (5 months)
Greater Brasilia

- Developed an IoT system for temperature monitoring and remote cooling control, preventing costly data center overheating; supported oversight of maintenance, IT, and telecom contracts (Python, MQTT, Embedded Systems)

Antal Bejczy Center for Intelligent Robotics
Junior Researcher
October 2014 - August 2015 (11 months)
Budapeste

- Developed Python software using the NAO SDK to control a humanoid robot, implementing visual pattern recognition to detect markers and trigger responsive movements (Python, NAO SDK, Computer Vision, Robotics).
- Developed software to train and guide students in laparoscopic procedures using a robotic arm with a phantom abdomen, enhancing surgical skills in a simulated environment (Python, OpenCV, PyQt)

Universidade de Brasília
Teaching Assistant
January 2013 - July 2014 (1 year 7 months)
Greater Brasilia

- Assisted in three undergraduate courses: Computer Vision Principles (30h), Algorithms & Data Structures (60h), and Experimental Physics 3 (60h)

CAPES - Coordenação de Aperfeiçoamento de Pessoal de Nível Superior
Young Talents for Science
April 2012 - April 2013 (1 year 1 month)
Greater Brasilia

- First hands-on experience in software prototyping and collaborative coding while studying the scientific method and hypothesis testing using the QuickSort algorithm and implementation of a Poker game (C, C++)

Education

Universite de Lorraine

Doctor of Philosophy - PhD, Automatique, Traitement du signal et des images, Génie informatique · (September 2022 - October 2025)

Faculty of Engineering, University of Debrecen

Master's degree, Mechatronics, Robotics, and Automation Engineering · (2020 - 2022)

Universidade de Brasília

Bachelor's degree, Mechatronics, Robotics, and Automation Engineering · (2012 - 2018)

Óbuda University

Bachelor's degree, Mechatronics, Robotics, and Automation Engineering · (2014 - 2015)